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1) Simplify:

a) $4m^2n + 5mn^2 - 2m^2n + mn^2 - 3mn^2$

b) $3x^2 - 5x + 2 + 3x^2 - 7x - 12$

2) Write as a single power of 2:

a) $2^5 \times 2^3$ b) $2^6 \div 2^2$ c) $\left(2^3\right)^2$

3) Expand:

a) $3(x + 4)$ b) $5(2 - 3x)$ c) $6(2x - 5y)$

4) Write down the highest common factor of:

a) 24 and 16

b) $6x$ and $8x^2$

c) $4xy^2$ and $3xy$

5) Simplify:

a) $\frac{10x}{5}$ b) $\frac{20x}{2}$ c) $\frac{40x}{24}$

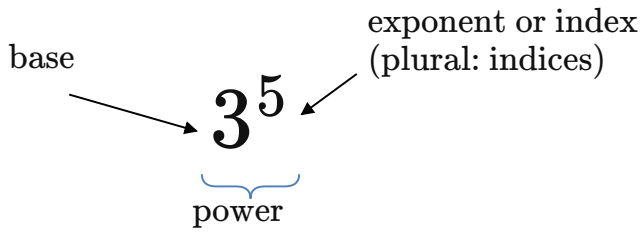
1: Basic Index Laws

2: Expand brackets

3: Factorise quadratics/cubics

4: Fractional/Negative Powers

5: Surds



$a^m \times a^n = a^{m+n}$
$a^m \div a^n = a^{m-n}$
$(a^m)^n = a^{mn}$
$(ab)^n = a^n b^n$

Some also refer to the 5 as the 'power'. But this would lead to unfortunate ambiguous phrases like "when we multiply two powers together, we add the powers" instead of "when we multiply two powers together, we add the indices".

Simplify $(a^3)^2 \times 2a^2$

Simplify $(4x^3y)^3$

Simplify $2x^2(3+5x) - x(4-x^2)$

Simplify $\frac{x^3-2x}{3x^2}$

split into
nt error
$+ \frac{a}{c}$

1 Simplify $\left(\frac{2a^5}{a^2}\right)^2 \times 3a$

2 Simplify $\frac{2x+x^5}{4x^3}$

3 Expand and simplify
$2x\left(3-x^2\right)-4x^3\left(3-x\right)$

4 Simplify $2^x \times 3^x$

2 Expand and simplify if possible:

o $x(3x^2-2x+5)$ p $7y^2(2-5y+3y^2)$ q $-2y^2(5-7y+3y^2)$

r $7(x-2)+3(x+4)-6(x-2)$ s $5x-3(4-2x)+6$

t $3x^2-x(3-4x)+7$ u $4x(x+3)-2x(3x-7)$ v $3x^2(2x+1)-5x^2(3x-4)$

3 Simplify these fractions:

d $\frac{8x^3+5x}{2x}$ **e** $\frac{7x^7+5x^2}{5x}$ **f** $\frac{9x^5-5x^3}{3x}$

Extension (Full Database: http://www.drfrostmaths.com/resources/resource.php?rid=268)

1 [MAT 2006 1A] Which of the following numbers is largest?

$$\left((2^3)^2\right)^3, \quad (2^3)^{(2^3)}, \quad 2^{((3^2)^3)}, \quad 2^{(3^{(2^3)})}$$

2 [MAT 2012 1B] Let $N = 2^k \times 4^m \times 8^n$ where k,m,n are positive whole numbers. Then N will definitely be a square number whenever:

k is even;

$k + n$ is odd;

k is odd but $m + n$ is even;

$k + n$ is even.

If you have ever been taught 'FOIL' to multiply brackets please purge it from your mind now – instead:

Multiply each term in the first bracket by each term in the second.

$$(x-y)(x+y-1)$$

$$= x^2 + xy - x - xy - y^2 + y$$

$$= x^2 - y^2 - x + y$$

Tip: My order is "first term in first brackets times each in second, then second term in first bracket times each in second, etc."

Tip: For more than 2 brackets, multiply two out each time to reduce the number of brackets by one.

1 Expand and simplify

$$(x+5)(x-2)(x+1)$$

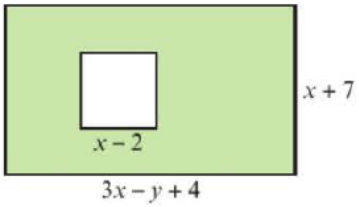
2 Expand and simplify:

$$2(x-3)(x-4)$$

3 Expand and simplify:

$$(2x-1)^3$$

3 The diagram shows a rectangle with a square cut out. The rectangle has length 3x-y+4 and width x+7. The square has length x-2. Find an expanded and simplified expression for the shaded area.



5 Given that $(2x+5y)(3x-y)(2x+y) = ax^3 + bx^2y + cxy^2 + dy^3$, where a,b,c,d are constants, find the values of a,b,c,d.

Extension (Full Database: http://www.drfrostmaths.com/resources/resource.php?rid=268)

1 [MAT 2002 1B] Of the following three alleged algebraic identities, at least one is wrong.

(i) $yz(z-y)+zx(x-z)+xy(y-x)$
$=(z-y)(x-z)(y-x)$

(ii) $yz(z-y)+zx(x-z)+xy(y-x)$
$=(z-y)(z-x)(y-x)$

(iii) $yz(x+y)+zx(z+x)+xy(y+x)$
$=(z+y)(z+x)(y+x)$

Which of the following statements are correct? Tick all that apply. (i), (ii), (iii).

2 [MAT 2007 1E] If x and n are integers then $(1-x)^n(2-x)^{2n}(3-x)^{3n}(4-x)^{4n}(5-x)^{5n}$ is:

negative when $n>5$ and $x<5$

negative when n is odd and $x>5$

negative when n is a multiple of 3 and $x>5$

negative when n is even and $x<5$

Informally, factorising is the opposite of expanding brackets.
More formally, a factorised expression is one which is expressed as **a product of expressions**.

$x(x+1)(x+2)$ ➡ Factorised as it is the product of 3 linear factors, x, $x+1$ and $x+2$.

Note: A linear expression is of the form $ax+b$. It is called linear because plotting $y=ax+b$ would form a straight line.

$x(x+1)+(x-1)(x+1)$ ➡ Not factorised because the outer-most operation is a sum, not a product.

Basic Examples:

$x^3+x^2 = \mathbf{x^2}(\mathbf{x+1})$
$4x-8xy = \mathbf{4x}(\mathbf{1-2y})$

Recap:

We find two numbers which multiply to give the coefficient of x and multiply to give the constant term.

$$\oplus \qquad \otimes$$

$$x^2 - 5x - 14 =$$

Note: The *coefficient* of a term is the constant on front of it, e.g. the coefficient of $4x^2$ is 4.

But what if the coefficient of x^2 is not 1?

$$2x^2 + 5x - 12 =$$

The easiest way is to use your common sense to guess the brackets. What multiplies to give the $2x^2$? What multiplies to give the constant term of -12?

Or you can 'split the middle term' (don't be embarrassed if you've forgotten how t

$x^2 + 5x - 12$

$\oplus 5$

$\otimes -$

$= 2$

$= 2$

$= ($

STEP 1: Find two numbers which add to give the middle number and multiply to give the first times last.

STEP 2: Split the middle term.

STEP 3: Factorise first half and second half ensuring bracket is duplicated..

STEP 4: Factorise out bracket.

Difference of two squares:

$$4x^2 - 9 =$$

Using multiple factorisations:

$x^3 - x$

$=x$

$=x($

Tip: Always look for a common factor first before using other factorisation techniques.

$x^3 + 3x^2 + 2x$

$=x$

$=x$

1 Factorise completely:
$6x^2 + x - 2$

2 Factorise completely:
$x^3 - 7x^2 + 12x$

Factorise completely:
$x^4 - 1$

Factorise completely:
$x^3 - 1$

4 Factorise completely $x^4 - y^4$.

5 Factorise completely $6x^3 + 7x^2 - 5x$.

Challenge: Write $4x^4 - 13x^2 + 9$as the product of four linear factors

$a^0 = 1$
$a^{\frac{1}{m}} = \sqrt[m]{a}$
$a^{\frac{n}{m}} = \left(\sqrt[m]{a}\right)^n$
$a^{-m} = \frac{1}{a^m}$

Note: $\sqrt{9}$ only means the positive square root of 9, i.e. 3 not -3. Otherwise, what would be the point of the $\pm$ in the quadratic formula before the $\sqrt{b^2 - 4ac}$?

Prove that $x^{\frac{1}{2}} = \sqrt{x}$

Evaluate $27^{-\frac{1}{3}}$

Evaluate $32^{\frac{2}{5}}$

Simplify $\left(\frac{1}{9}x^6y\right)^{\frac{1}{2}}$

Evaluate $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$

If $b = \frac{1}{9}a^2$, determine $3b^{-2}$ in the form kb^n where k, n are constants.

If $9\sqrt{3} = 3^k$, determine the value of k.

[Edexcel IGCSE May14-4H]
Given that

$$\left(2^{\frac{1}{2}}\right)^{n} = \frac{2^{x}}{8^{y}}$$

Express n in terms of x and y.

[Edexcel IAL C12 Jan 2019 Q2c] Given that $y = 2^{x}$, express the following in terms of y.

$$\frac{1}{4^{2x-3}}$$

Write your expression in its simplest form.

Given $2^{x} \times 4^{y} = \frac{1}{2\sqrt{2}}$ express y as a function of x. (3)

4 a Find the value of $81^{\frac{1}{4}}$.
b Simplify $x(2x^{-\frac{1}{3}})^4$.

5 Given that $y = \frac{1}{8}x^3$ express each of the following in the form kx^n, where k and n are constants.

a $y^{\frac{1}{3}}$ **b** $\frac{1}{2}y^{-2}$

Challenge

[MAT 2007 1A] Let r and s be integers. Then $\frac{6^{r+s}\times 12^{r-s}}{8^r\times 9^{r+2s}}$ is an integer if $r+s\leq 0$; $s\leq 0$; $r\leq 0$; $r\geq s$.

Recap:

A surd is a root of a number that does not simplify to a rational number.

Laws:
$\sqrt{a}\times\sqrt{b}=\sqrt{ab}$
$\frac{\sqrt{a}}{\sqrt{b}}=\sqrt{\frac{a}{b}}$

Note: A *rational* number is any which can be expressed as $\frac{a}{b}$ where a, b are integers. $\frac{2}{3}$ and $\frac{4}{1}=4$ are rational numbers, but π and $\sqrt{2}$ are not.

$\sqrt{3}\times 2 = \mathbf{2}$
$3\sqrt{5}\times 2\sqrt{5} = \mathbf{30}$
$\sqrt{8}=\sqrt{4}\sqrt{2} = \mathbf{2}\sqrt{}$
$\sqrt{12}+\sqrt{27} = \mathbf{2}\sqrt{3}$
$\left(\sqrt{8}+1\right)\left(\sqrt{2}-3\right) = \sqrt{16}$
$= \mathbf{4}$
$= \mathbf{1}$

3 Simplify $\sqrt{75} - \sqrt{12}$ giving your answer in the form $a\sqrt{3}$, where a is an integer.

Extension

1 [SMC 2014 Q24] Which of the following is smallest?

$10 - 3\sqrt{11}$; $8 - 3\sqrt{7}$; $5 - 2\sqrt{6}$; $9 - 4\sqrt{5}$; $7 - 4\sqrt{3}$.

2 [SMC 2012 Q21] Which of the following numbers does not have a square root in the form $x + y\sqrt{2}$, where x and y are positive integers?

$17 + 12\sqrt{2}$; $22 + 12\sqrt{2}$; $38 + 12\sqrt{2}$; $54 + 12\sqrt{2}$; $73 + 12\sqrt{2}$.

Here's a surd. What could we multiply it by such that it's no longer an irrational number?

$$\sqrt{5} \times \square = \square$$

$$\frac{1}{\sqrt{2}} \times \square = \square$$

In this fraction, the denominator is irrational. '**Rationalising the denominator**' means making the denominator a rational number.

What could we multiply this fraction by to both rationalise the denominator, but leave the value of the fraction unchanged?

Side Note: There's two reasons why we might want to do this:

1. For aesthetic reasons, it makes more sense to say "half of root 2" rather than "one root two-th of 1". It's nice to divide by something whole!
2. It makes it easier for us to add expressions involving surds.

Test Your Understanding:

$\frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$

$\frac{6}{\sqrt{3}} = \frac{6\sqrt{3}}{3} = 2\sqrt{3}$

$\frac{7}{\sqrt{7}} = \frac{7\sqrt{7}}{7} = \sqrt{7}$

$\frac{15}{\sqrt{5}} + \sqrt{5} = 4\sqrt{5}$

$\frac{12}{\sqrt{3}} = \sqrt{3}$

$\frac{2}{\sqrt{6}} = \frac{\sqrt{6}}{3}$

$\frac{4\sqrt{2}}{\sqrt{8}} = 2$

You've seen 'rationalising a denominator', the idea being that we don't like to divide things by an irrational number.

But what do we multiply the top and bottom by if we have a more complicated denominator?

$$\frac{1}{\sqrt{2}+1} = \frac{\quad}{\quad}$$

We basically use the same expression but with the sign reversed (this is known as the *conjugate*). That way, we obtain the difference of two squares. Since $(a+b)(a-b) = a^2 - b^2$, any surds will be squared and thus we'll end up with no surds in the denominator.

$\frac{3}{\sqrt{6}-2} \times$ [ ] $=$ [ ]

You can explicitly expand out $\left(\sqrt{6}-2\right)\left(\sqrt{6}+2\right)$ in the denominator, but remember that $(a-b)(a+b)=a^2-b^2$ so we can mentally obtain $6-4=2$
Just remember: 'difference of two squares'!

$\frac{4}{\sqrt{3}+1} \times$ [ ] $=$ [ ] $2\sqrt{}$ [ ]

$\frac{3\sqrt{2}+4}{5\sqrt{2}-7} \times \frac{5\sqrt{2}+7}{5\sqrt{2}+7}$ [ ] $\frac{\sqrt{2}+}{1}$ $=$ [ ]

Rationalise the denominator and simplify $\frac{4}{\sqrt{5}-2}$

[ ]

Rationalise the denominator and simplify

$$\frac{2\sqrt{3}-1}{3\sqrt{3}+1}$$

[ ]

AQA IGCSE FM June 2013 Paper 1

Solve $y\left(\sqrt{3}-1\right)=8$

Give your answer in the form $a+b\sqrt{3}$ where a and b are integers.

[ ]

This is not in the textbook, but a common type of question in exams, often asked in conjunction with differentiation (a later chapter).

Express $\frac{(2+\sqrt{x})^2}{x}$ as powers of x.

Test Your Understanding

Express $\frac{(x+\sqrt{x})^2}{2\sqrt{x}}$ as powers of x.

or alternatively: (not in textbook)

1 Rationalise the denominator and simplify the following:

a $\frac{1}{\sqrt{5}+2} = \sqrt{5}-2$

b $\frac{\sqrt{3}}{\sqrt{3}-1} = \frac{3+\sqrt{3}}{2}$

c $\frac{\sqrt{5}+1}{\sqrt{5}-2} = 7+3\sqrt{5}$

d $\frac{2\sqrt{3}-1}{3\sqrt{3}+4} = 2-\sqrt{3}$

e $\frac{5\sqrt{5}-2}{2\sqrt{5}-3} = 4+\sqrt{5}$

2 Expand and simplify:
$\left(\sqrt{5}+3\right)\left(\sqrt{5}-2\right)\left(\sqrt{5}+1\right) =$

3 Rationalise the denominator, giving your answer in the form $a+b\sqrt{3}$.
$\frac{3\sqrt{3}+7}{3\sqrt{3}-5} =$

4 Solve $x\left(4-\sqrt{6}\right) = 10$ giving your answer in the form $a+b\sqrt{6}$.
$x = \frac{10}{4-\sqrt{6}} = 4+\sqrt{6}$

Solv
$y = \frac{3+\sqrt{2}}{1+\sqrt{2}} = 2\sqrt{2}-1$

5 Simplify:
$\frac{\sqrt{a+1}-\sqrt{a}}{\sqrt{a+1}+\sqrt{a}}$

6

Solve $\frac{\sqrt[4]{9}}{\sqrt[5]{27}} = \sqrt[x]{3}$